# TUGAS P7 - PEMROGRAMAN

# Variabel Array



Disusun Oleh :


| | | |
|---|---|---|
| NAMA | : | Muhammad Aditya Rahmansyah |
| NRP | : | 3226600122 |
| KELAS | : | ST.r Tekkom A |
| MATA KULIAH | : | Pemrograman |
| DOSEN | : | Ir. Sigit Wasista, M.Kom. |
| ASSISTEN DOSEN | : | Rofiulloh Mu'azin Al-insof, A.Md. |
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DEPARTEMEN TEKNIK INFORMATIKA DAN KOMPUTER
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA
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Tugas 7.1 Tampilkan Data

Code :

```c
#include <stdio.h>

#define N 8

int main()
{
    int data[N] = {7, 8, 4, 2, 9, 3, 1, 6};
    int urut[N];
    int i, j, tmp, terbesar, jumlah = 0;
    double rata2, tengah;


    printf("Data (index 1..%d):\n", N);
    for (i = 0; i < N; i++)
        printf("data[%d] = %d\n", i + 1, data[i]);


    printf("\nAngka genap : ");
    for (i = 0; i < N; i++)
        if (data[i] % 2 == 0)
            printf("%d ", data[i]);
    printf("\n");


    terbesar = data[0];
    for (i = 1; i < N; i++)
        if (data[i] > terbesar)
            terbesar = data[i];
    printf("Angka terbesar : %d\n", terbesar);

    for (i = 0; i < N; i++)
        urut[i] = data[i];
    for (i = 0; i < N - 1; i++)
        for (j = 0; j < N - 1 - i; j++)
            if (urut[j] > urut[j + 1]) {
                tmp = urut[j];
                urut[j] = urut[j + 1];
                urut[j + 1] = tmp;
            }
    printf("Urut dari kecil : ");
    for (i = 0; i < N; i++)
        printf("%d ", urut[i]);
    printf("\n");


    for (i = 0; i < N; i++)
        jumlah += data[i];
    rata2 = (double)jumlah / N;
    printf("Rata-rata : %.2f\n", rata2);


    if (N % 2 == 1)
        tengah = urut[N / 2];
    else
        tengah = (urut[N / 2 - 1] + urut[N / 2]) / 2.0;
    printf("Nilai tengah (median) : %.2f\n", tengah);

    return 0;
}
```

Output :

```
Data (index 1..8):
data[1] = 7
data[2] = 8
data[3] = 4
data[4] = 2
data[5] = 9
data[6] = 3
data[7] = 1
data[8] = 6

Angka genap : 8 4 2 6
Angka terbesar : 9
Urut dari kecil : 1 2 3 4 6 7 8 9
Rata-rata : 5.00
Nilai tengah (median) : 5.00
```

Tugas 7.2 Urutan naik

Code :

```c
#include <stdio.h>

#define MAXDATA 100

int main()
{
    int data[MAXDATA];
    int n, i, j, tmp;

    do {
        printf("Jumlah data (minimal 5, maksimal %d) : ", MAXDATA);
        scanf("%d", &n);
        if (n < 5 || n > MAXDATA)
            printf("Jumlah data tidak valid!\n");
    } while (n < 5 || n > MAXDATA);

    for (i = 0; i < n; i++) {
        printf("Data ke-%d : ", i + 1);
        scanf("%d", &data[i]);
    }

    for (i = 0; i < n - 1; i++)
        for (j = 0; j < n - 1 - i; j++)
            if (data[j] > data[j + 1]) {
                tmp = data[j];
                data[j] = data[j + 1];
                data[j + 1] = tmp;
            }

    printf("\nData setelah diurutkan (naik) :\n");
    for (i = 0; i < n; i++)
        printf("%d ", data[i]);
    printf("\n");

    return 0;
}
```

Output :

```
Jumlah data (minimal 5, maksimal 100) : 5
Data ke-1 : 90
Data ke-2 : 23
Data ke-3 : 56
Data ke-4 : 78
Data ke-5 : 35

Data setelah diurutkan (naik) :
23 35 56 78 90
```

Tugas 7.3 Bilangan prima

Code :

```c
#include <stdio.h>

#define MAX 46349

static int P[MAX + 1];

int main()
{
    long int i, j;
    long int n;
    int cacah = 0;

    do {
        printf("Tampilkan bilangan prima antara 1 sampai n (n <= %d) : ", MAX);
        scanf("%ld", &n);
    } while (n < 2 || n > MAX);

    for (i = 2; i <= n; i++) {
        if (P[i] == 0)
            for (j = i * i; j <= n; j += i)
                P[j] = 1;
    }

    printf("\nBilangan prima 1 sampai %ld :\n", n);
    for (i = 2; i <= n; i++) {
        if (P[i] == 0) {
            printf("%8ld", i);
            if (++cacah % 10 == 0)
                printf("\n");
        }
    }
    printf("\n\nJumlah bilangan prima = %d\n", cacah);

    return 0;
}
```

Output :

```
Tampilkan bilangan prima antara 1 sampai n (n <= 46349) : 56

Bilangan prima 1 sampai 56 :
       2       3       5       7      11      13      17      19      23      29
      31      37      41      43      47      53

Jumlah bilangan prima = 16
```

Tugas 7.4 Bujursangkar Ajaib

Code :

```c
#include <stdio.h>

int main()
{
    int ajaib[19][19] = {{0}};
    int ordo, tengah, baris, kolom, angka, b, k;
    int nb, nk, total, sama = 1, jml;

    do {
        printf("Ordo (ganjil, 3 s/d 19) = ");
        scanf("%d", &ordo);
    } while (!(ordo >= 3 && ordo <= 19) || ordo % 2 == 0);

    tengah = ordo / 2;
    baris = 0;
    kolom = tengah;

    for (angka = 1; angka <= ordo * ordo; angka++) {
        ajaib[baris][kolom] = angka;

        nb = baris - 1;
        nk = kolom + 1;
        if (nb < 0)     nb = ordo - 1;
        if (nk == ordo) nk = 0;

        if (ajaib[nb][nk] != 0) {
            nb = baris + 1;
            nk = kolom;
        }
        baris = nb;
        kolom = nk;
    }

    printf("\nBujursangkar ajaib ordo %d x %d:\n\n", ordo, ordo);
```

```c
    for (b = 0; b < ordo; b++) {
        for (k = 0; k < ordo; k++)
            printf("%5d", ajaib[b][k]);
        printf("\n");
    }

    jml = ordo * (ordo * ordo + 1) / 2;
    for (b = 0; b < ordo; b++) {
        total = 0;
        for (k = 0; k < ordo; k++) total += ajaib[b][k];
        if (total != jml) sama = 0;
    }
    for (k = 0; k < ordo; k++) {
        total = 0;
        for (b = 0; b < ordo; b++) total += ajaib[b][k];
        if (total != jml) sama = 0;
    }
    printf("\nJumlah tiap baris/kolom = %d (%s)\n", jml, sama ? "terverifikasi" : "tidak sama");

    return 0;
}
```

Output :

```
Ordo (ganjil, 3 s/d 19) = 3

Bujursangkar ajaib ordo 3 x 3:

    8    1    6
    3    5    7
    4    9    2

Jumlah tiap baris/kolom = 15 (terverifikasi)
```

Tugas 7.5 Menghitung nama hari

Code :

```c
#include <stdio.h>

int main()
{
    int kodeBulan1[12]  = {1, 4, 4, 0, 2, 5, 0, 3, 6, 1, 4, 6};
    int kodeBulan1K[12] = {0, 3, 4, 0, 2, 5, 0, 3, 6, 1, 4, 6};
    char *namaHari[7] = {"Sabtu", "Minggu", "Senin", "Selasa", "Rabu", "Kamis", "Jumat"};

    int hari2[12]   = {6, 3, 3, 0, 5, 2, 0, 4, 1, 6, 3, 1};
    int hari2K[12]  = {0, 4, 3, 0, 5, 2, 0, 4, 1, 6, 3, 1};
    int pasar2[12]  = {0, 4, 1, 0, 0, 4, 4, 3, 2, 2, 1, 1};
    int pasar2K[12] = {1, 0, 1, 0, 0, 4, 4, 3, 2, 2, 1, 1};
    char *namaPasaran[5] = {"Legi", "Pahing", "Pon", "Wage", "Kliwon"};

    int tgl, bln, thn, kabisat, kode, kode2, pas;

    printf("Masukkan tanggal (dd-mm-yy) : ");
    scanf("%d-%d-%d", &tgl, &bln, &thn);

    if (thn >= 1900) thn -= 1900;
    if (tgl < 1 || tgl > 31 || bln < 1 || bln > 12 || thn < 0 || thn > 99) {
        printf("Tanggal tidak valid (tahun harus 1900-1999)!\n");
        return 1;
    }

    kabisat = (thn % 4 == 0 && thn != 0);

    kode = (tgl + (kabisat ? kodeBulan1K[bln - 1] : kodeBulan1[bln - 1])
            + thn + thn / 4) % 7;
    printf("Tanggal tersebut adalah hari %s.\n", namaHari[kode]);

    kode2 = ((tgl + thn + thn / 4) - (kabisat ? hari2K[bln - 1] : hari2[bln - 1])) % 7;
    if (kode2 < 0) kode2 += 7;
    pas = (tgl + thn / 4 - (kabisat ? pasar2K[bln - 1] : pasar2[bln - 1])) % 5;
    if (pas < 0) pas += 5;
```

```c
    printf("Formula II : hari %s %s.\n", namaHari[kode2], namaPasaran[pas]);

    return 0;
}
```

Output :

```
Masukkan tanggal (dd-mm-yy) : 16-03-1999
Tanggal tersebut adalah hari Selasa.
Formula II : hari Selasa Kliwon.
```

Tugas 7.6 Menghitung nama hari

Code :

```c
#include <stdio.h>

static const int kode[28][12] = {
    {5,1,2,5,0,3,5,1,4,6,2,4},
    {0,3,3,6,1,4,6,2,5,0,3,5},
    {1,4,4,0,2,5,0,3,6,1,4,6},
    {2,5,5,1,3,6,1,4,0,2,5,0},
    {3,6,0,3,5,1,3,6,2,4,0,2},
    {5,1,1,4,6,2,4,0,3,5,1,3},
    {6,2,2,5,0,3,5,1,4,6,2,4},
    {0,3,3,6,1,4,6,2,5,0,3,5},
    {1,4,5,1,3,6,1,4,0,2,5,0},
    {3,6,6,2,4,0,2,5,1,3,6,1},
    {4,0,0,3,5,1,3,6,2,4,0,2},
    {5,1,1,4,6,2,4,0,3,5,1,3},
    {6,2,3,6,1,4,6,2,5,0,3,5},
    {1,4,4,0,2,5,0,3,6,1,4,6},
    {2,5,5,1,3,6,1,4,0,2,5,0},
    {3,6,6,2,4,0,2,5,1,3,6,1},
    {4,0,1,4,6,2,4,0,3,5,1,3},
    {6,2,2,5,0,3,5,1,4,6,2,4},
    {0,3,3,6,1,4,6,2,5,0,3,5},
    {1,4,4,0,2,5,0,3,6,1,4,6},
    {2,5,6,2,4,0,2,5,1,3,6,1},
    {4,0,0,3,5,1,3,6,2,4,0,2},
    {5,1,1,4,6,2,4,0,3,5,1,3},
    {6,2,2,5,0,3,5,1,4,6,2,4},
    {0,3,4,0,2,5,0,3,6,1,4,6},
    {2,5,5,1,3,6,1,4,0,2,5,0},
    {3,6,6,2,4,0,2,5,1,3,6,1},
    {4,0,0,3,5,1,3,6,2,4,0,2}
};
```

```c
int main()
{
    char *namaHari[7] = {"Sabtu", "Minggu", "Senin", "Selasa", "Rabu", "Kamis", "Jumat"};
    int hariBulan[12] = {31,28,31,30,31,30,31,31,30,31,30,31};
    int tgl, bln, thn, maks, kd, neptu;

    printf("Masukkan tanggal (dd-mm-yyyy) : ");
    scanf("%d-%d-%d", &tgl, &bln, &thn);

    if (thn < 1901 || thn > 2099 || bln < 1 || bln > 12) {
        printf("Input tidak valid (tahun 1901-2099, bulan 1-12)!\n");
        return 1;
    }
    maks = hariBulan[bln - 1];
    if (bln == 2 && thn % 4 == 0) maks = 29;
    if (tgl < 1 || tgl > maks) {
        printf("Tanggal tidak valid!\n");
        return 1;
    }

    kd = kode[thn % 28][bln - 1];
    neptu = tgl + kd;
    printf("Kode = %d, tanggal + kode = %d + %d = %d\n", kd, tgl, kd, neptu);
    printf("Hari %02d-%02d-%d adalah hari %s\n", tgl, bln, thn, namaHari[neptu % 7]);

    return 0;
}
```

Output :

```
Masukkan tanggal (dd-mm-yyyy) : 16-03-1999
Kode = 1, tanggal + kode = 16 + 1 = 17
Hari 16-03-1999 adalah hari Selasa
```

Tugas 7.7 Menghitung nama pasaran

Code :

```c
#include <stdio.h>

static const int kode[20][12] = {
    {2,3,2,3,3,4,4,0,1,1,2,2},
    {3,4,2,3,3,4,4,0,1,1,2,2},
    {3,4,2,3,3,4,4,0,1,1,2,2},
    {3,4,2,3,3,4,4,0,1,1,2,2},
    {3,4,3,4,4,0,0,1,2,2,3,3},
    {4,0,3,4,4,0,0,1,2,2,3,3},
    {4,0,3,4,4,0,0,1,2,2,3,3},
    {4,0,3,4,4,0,0,1,2,2,3,3},
    {4,0,4,0,0,1,1,2,3,3,4,4},
    {0,1,4,0,0,1,1,2,3,3,4,4},
    {0,1,4,0,0,1,1,2,3,3,4,4},
    {0,1,4,0,0,1,1,2,3,3,4,4},
    {0,1,0,1,1,2,2,3,4,4,0,0},
    {1,2,0,1,1,2,2,3,4,4,0,0},
    {1,2,0,1,1,2,2,3,4,4,0,0},
    {1,2,0,1,1,2,2,3,4,4,0,0},
    {1,2,1,2,2,3,3,4,0,0,1,1},
    {2,3,1,2,2,3,3,4,0,0,1,1},
    {2,3,1,2,2,3,3,4,0,0,1,1},
    {2,3,1,2,2,3,3,4,0,0,1,1}
};
```

```c
int main()
{
    char *namaPasaran[5] = {"Pon", "Wage", "Kliwon", "Legi", "Pahing"};
    int hariBulan[12] = {31,28,31,30,31,30,31,31,30,31,30,31};
    int tgl, bln, thn, maks, kd, total;

    printf("Masukkan tanggal (dd-mm-yyyy) : ");
    scanf("%d-%d-%d", &tgl, &bln, &thn);

    if (thn < 1901 || thn > 2099 || bln < 1 || bln > 12) {
        printf("Input tidak valid (tahun 1901-2099, bulan 1-12)!\n");
        return 1;
    }
    maks = hariBulan[bln - 1];
    if (bln == 2 && thn % 4 == 0) maks = 29;
    if (tgl < 1 || tgl > maks) {
        printf("Tanggal tidak valid!\n");
        return 1;
    }

    kd = kode[thn % 20][bln - 1];
    total = tgl + kd;
    printf("Kode = %d, tanggal + kode = %d + %d = %d\n", kd, tgl, kd, total);
    printf("Pasaran %02d-%02d-%d adalah %s\n", tgl, bln, thn, namaPasaran[total % 5]);

    return 0;
}
```

Output :

```
Masukkan tanggal (dd-mm-yyyy) : 16-03-1999
Kode = 1, tanggal + kode = 16 + 1 = 17
Pasaran 16-03-1999 adalah Kliwon
```